

Task #4

Q71. Assembly_Trusses.py

Description: This file handles the construction of the stiffness matrices and force vectors required for the global system of equations.

- **LocalStiffnessAssembly(bar):** Calculates the 4×4 element stiffness matrix for a single bar in global coordinates using its material properties and orientation.
- **LocalToGlobalIndexing(bar):** Maps the local degrees of freedom (0-3) of a bar to their corresponding indices in the global stiffness matrix.
- **AssembleStiffness(bars, n_matrix):** Iterates through all bars to compile their local contributions into a single global stiffness matrix.
- **DefineForces(nodes, n_matrix):** Populates the global force vector by aggregating external loads applied to each node.

Q72. Classes_Trusses.py

Description: This file defines the object-oriented structure of the program, representing the physical components of the truss.

- **Node Class:** Stores data for a single point in the truss, including its coordinates, applied forces, calculated displacements, and reaction forces.
- **Bar Class:** Represents a structural member connecting two nodes, storing material properties (E), geometric properties (A, I), and calculated internal results like axial load and stress.

Q73. DoFIndexing_Trusses.py

Description: This file manages the numbering and organization of the Degrees of Freedom (DoF) to separate known and unknown values.

- **SetNodeIdx(node, constraint_dir, dof_idx):** Assigns a specific global index to a node's X or Y degree of freedom.
- **EstablishGlobalDOFNum(nodes):** Organizes all nodal DoFs so that unknowns come first, followed by knowns (constrained values), and returns the counts for both.
- **StoreNodeDisplacements(nodes, d, n_unknowns):** Takes the solved displacement vector and maps those values back into the respective Node objects.

Q74. ImportCSVData_Trusses.py

Description: This file serves as the data ingestion layer, reading truss geometry and properties from an external CSV file.

- **LoadData(input_geometry, section_file, material_file):** Parses the CSV to create **Node** and **Bar** objects, links them together, and fetches section/material properties from database files.

Q75. Main_Trusses.py

Description: This is the primary execution script that coordinates the entire analysis workflow from data import to final results.

- **PlaneTrussStiffness(input_geometry):** The core wrapper function that calls the import, assembly, solver, and post-processing functions in the correct sequence.

Q76. SectionMaterialConverter.py

Description: This file acts as a utility to look up and assign engineering properties from external databases to the structural members.

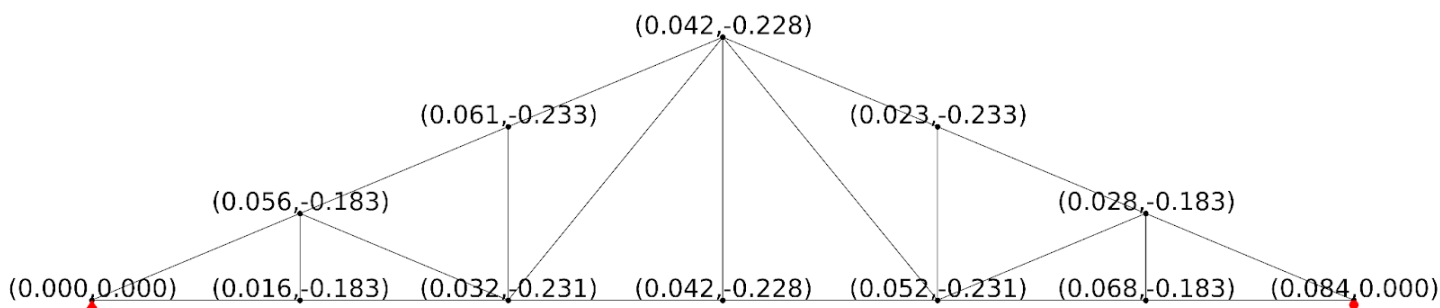
- **LoadMaterialData(bar, material, material_file):** Searches a material database to assign the Modulus of Elasticity (E) and Shear Modulus (G) to a bar.
- **LoadSectionData(bar, section, section_file):** Searches a shape database (like AISC) to assign Area (A), Moments of Inertia (I), and self-weight to a bar.

Q77. Solver_Trusses.py

Description: This file contains the mathematical operations to solve the linear system and calculate secondary engineering results.

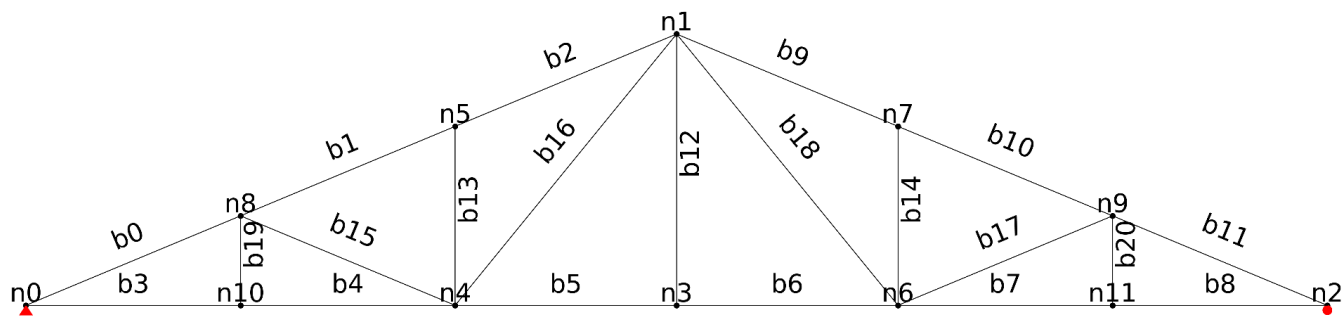
- **ComputeDisplacements(K, F, n_unknowns):** Uses linear algebra to solve the $Kd=F$ system for the unknown nodal displacements.
- **PostprocessReactions(K, d, F, n_unknowns, nodes):** Calculates the reaction forces at supported nodes using the solved displacements and the stiffness matrix.
- **ComputeMemberForces(bars):** Intended to calculate the internal axial force for each bar.
- **ComputeNormalStresses(bars):** Intended to calculate the axial stress based on internal force and cross-sectional area.
- **ComputeBucklingLoad(bars):** Intended to calculate the Euler buckling limit for members in compression.

Q82 Nodes w/ Displacements:

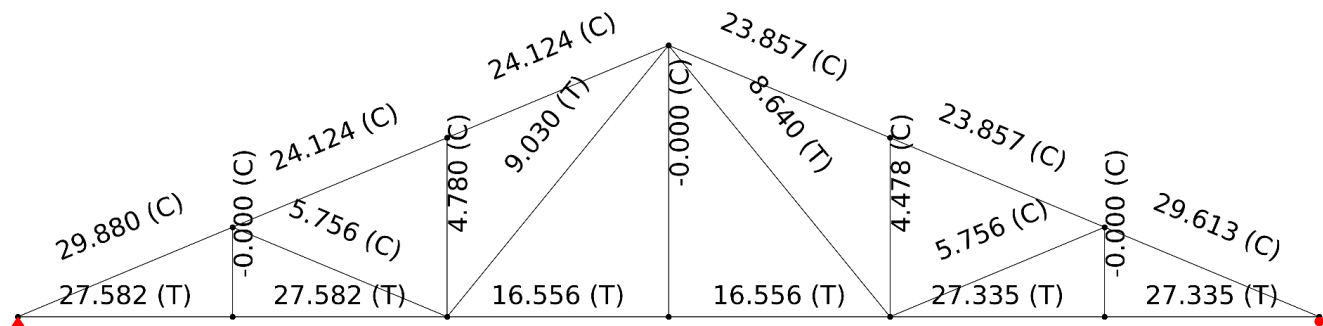


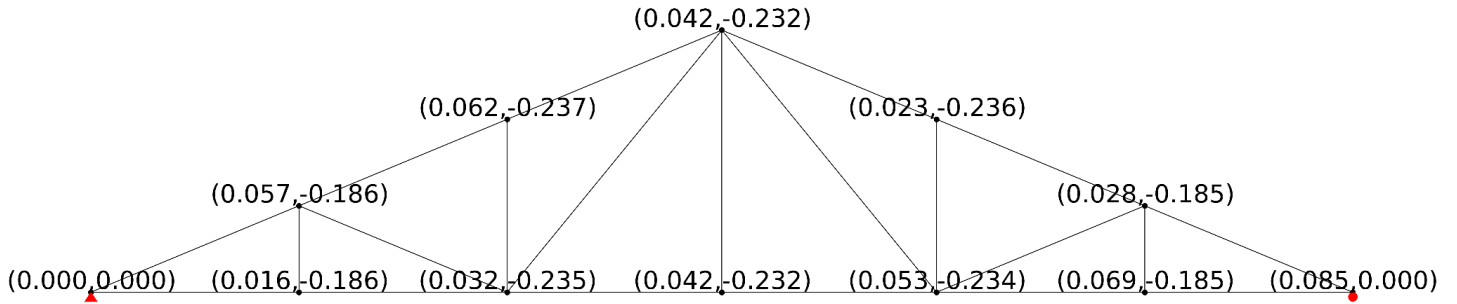
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Q86. Nodes and Bars

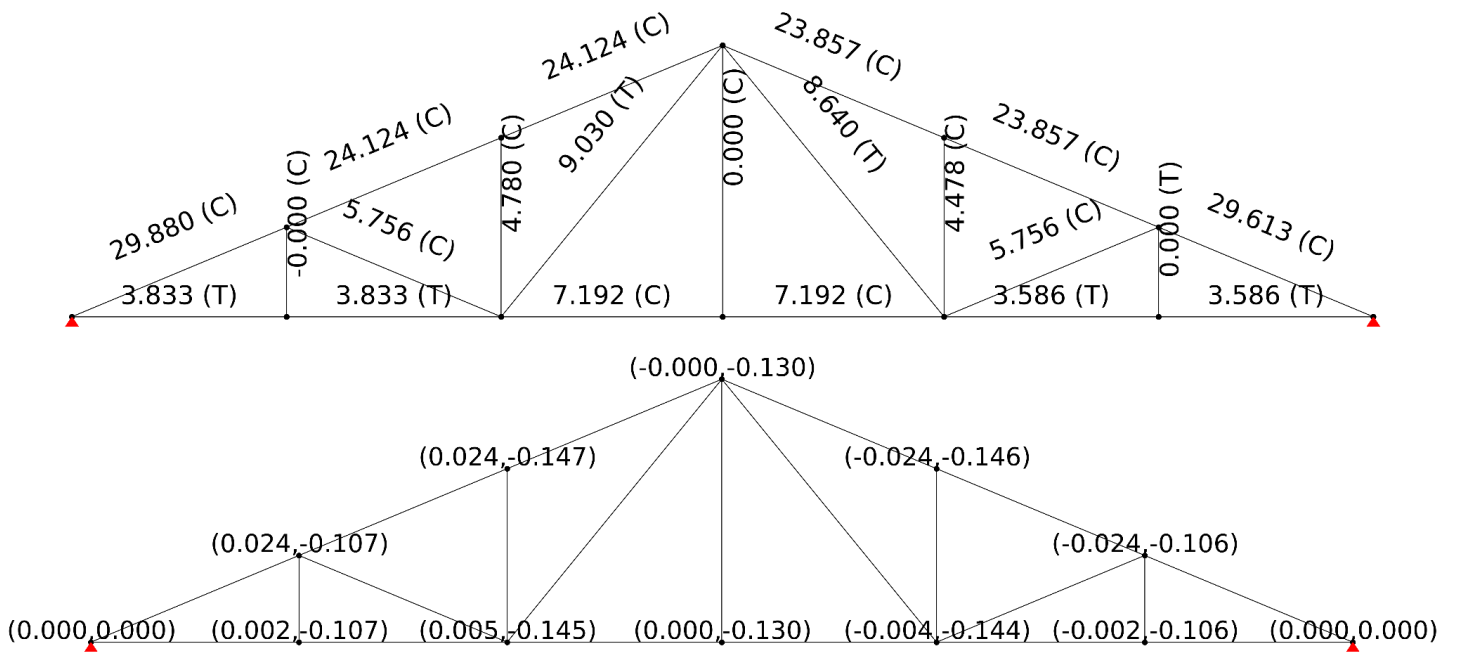


Q87.
No Wind, Simply Supported

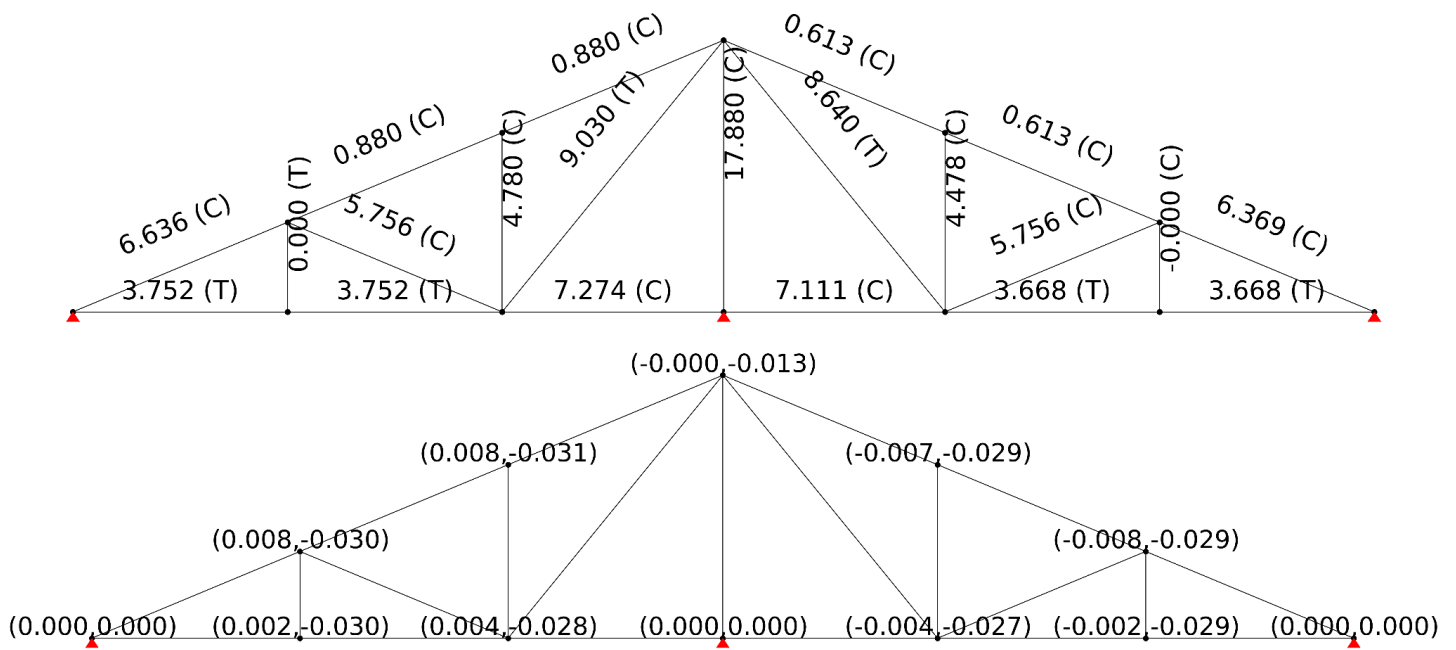




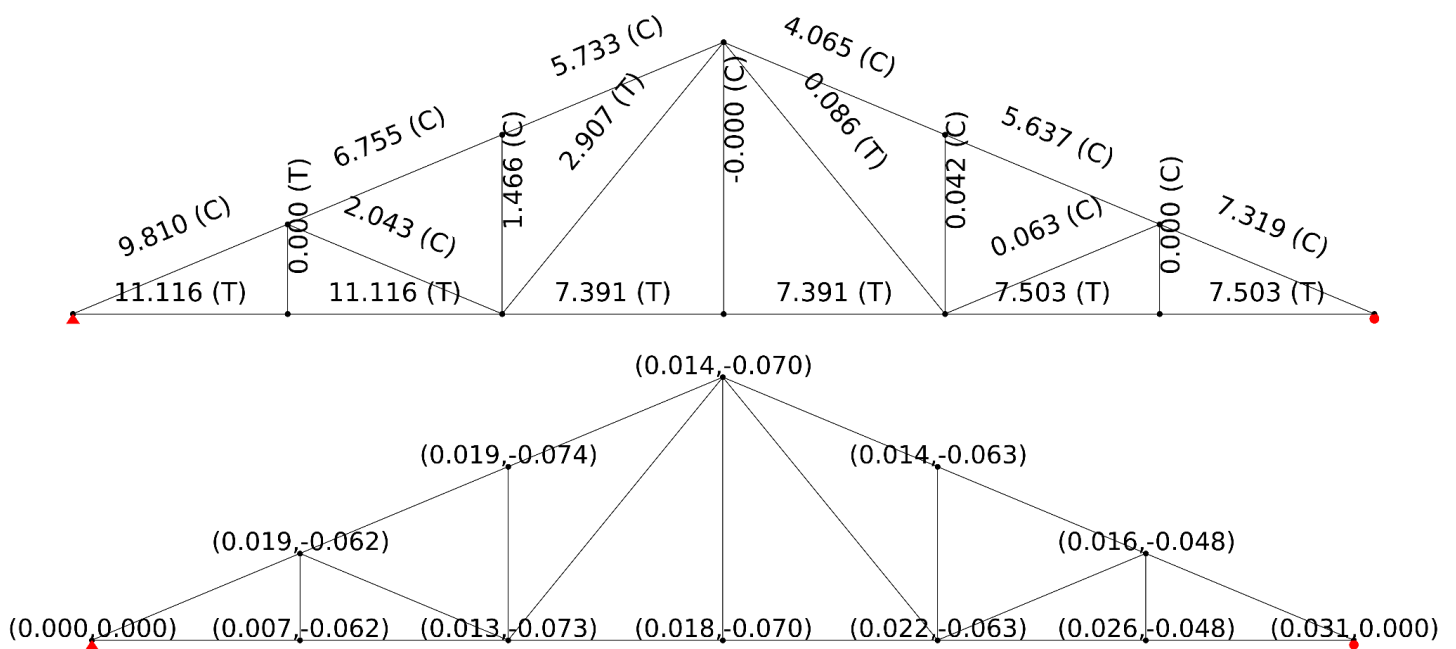
Q88. No Wind, Two Pins



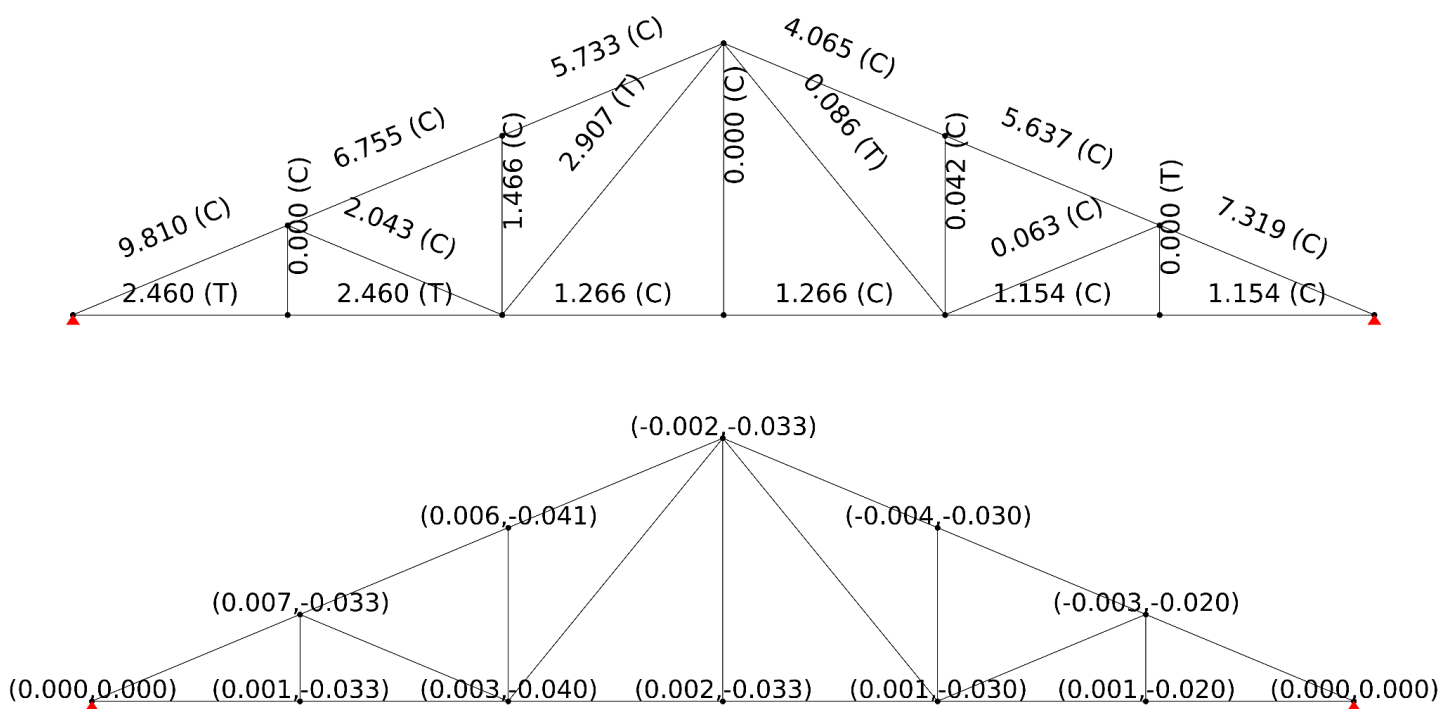
Q89. No Wind, Three Pins



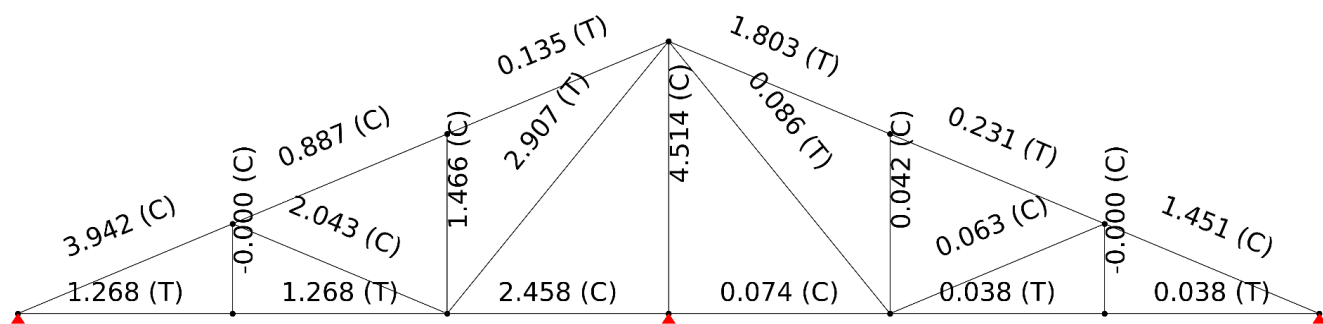
Q90. Wind, Simply Supported

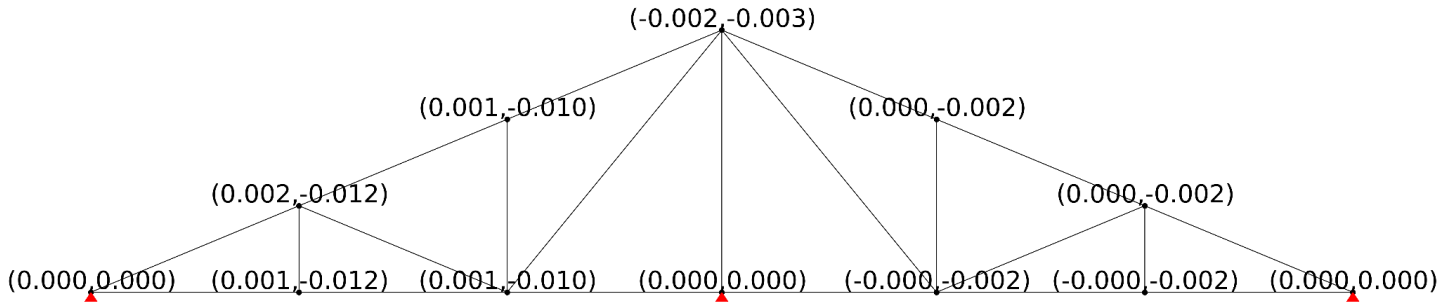


Q91. Wind, Two Pins



Q92. Wind, Three Pins





Q93 Commentary.

1. No Wind, Simply Supported

In this baseline configuration, the truss exhibits the highest magnitude of displacement and internal stress.

- **Maximum Displacement:** The central nodes show vertical deflections as high as -0.233 to -0.235 units.
- **Axial Loads:** Member loads are substantial, with some compression members reaching 29.880 (C) and tension members reaching 27.582 (T).

2. No Wind, Two & Three Pins

Adding pin supports (which restrict horizontal movement) significantly reduces overall deflection and redistributes the internal forces.

- **Two Pins:** The vertical displacement at the peak (node n1) is reduced from approximately -0.23 to -0.130 . Axial loads in the bottom chord are lowered to roughly 3.833 (T).
- **Three Pins:** Adding a third pin support further stiffens the structure. Peak vertical displacement drops drastically to -0.013 , and axial loads are further minimized across several members.

3. Wind Load Effects

When wind loading is introduced, the symmetry of the internal forces is disrupted.

Axial Load Distribution (With Wind)

Under wind conditions with three pin supports:

- **Asymmetry:** The loads are no longer mirrored perfectly. For instance, some members show compression of 3.942 (C) while the corresponding member on the other side might be at 1.451 (C).
- **Specific Member Reactions:** High compression is noted in the central vertical or diagonal members, such as 4.514 (C) in bar b12.
- **Minimum Displacement:** The displacements remain very low due to the three-pin constraint, with the peak node (n1) moving only -0.003 vertically.